

**Ascham School
Trial Higher School Certificate
Mathematics 4 unit**

July 1999

Time allowed: 3 hours

Instructions to Students

1. Attempt all questions
2. All questions are of equal value
3. Answer each question in a separate booklet
4. Marks may not be awarded for careless or badly arranged work
5. Approved calculators may be used
6. Table of Standard Integrals are provided

Question 1 (15 marks)

a) Find $\int 7x\sqrt{4x^2 - 3} dx$ 2

Evaluate the following definite integrals

(b) (i) $\int_0^{\sqrt{2}} \sqrt{4 - x^2} dx$ 3

(ii) $\int_0^{\pi} x \sin x dx$ 3

(iii) $\int_2^4 \frac{dx}{x^2 - 4x + 8}$ 3

(iv) $\int_{-1}^1 \frac{4 + x^2}{4 - x^2} dx$ 4

Question 2 (15 marks) START A NEW BOOKLET

a) (i) Solve $x^2 - 3ix + 4 = 0$ 2

(ii) Express $\sqrt{12 - 5i}$ in the form $a + ib$, where a, b are real 4

(iii) Find the locus of z , where $z = \frac{u - i}{u - 2}$ 5

α) If u is purely real

β) If u moves around a unit circle

(iv) Indicate on an Argand diagram the region in which both the following inequalities are satisfied. 4

$$|z - (3 + i)| \leq 3 \quad \text{and} \quad \frac{\pi}{4} \leq \arg[z - (1 + i)] \leq \frac{\pi}{2}$$

Question 3 (15 marks) START A NEW BOOKLET

a) Let $I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx$ where n is an integer and $n \geq 3$ 4

Show that $I_n + I_{n-2} = \frac{1}{n-1}$ and hence evaluate I_5

b) (i) If $u = \frac{1+i}{\sqrt{2}}$, show that $u^4 = -1$

(ii) On an Argand diagram illustrate the roots of the equation $z^4 = 1$ 11

(iii) On the same diagram illustrate the roots of the equation $z^4 = -1$

(iv) Hence or otherwise write down the solutions of the equation $z^8 - 1 = 0$

Question 4 (15 marks) START A NEW BOOKLET

- a) The roots of the polynomial $P(x) = 4x^3 - 12x^2 + 11x - 3$ are in arithmetic sequence. Solve $P(x) = 0$ over the real number system. 4
- b) (i) Prove that if $Q(x)$ is a polynomial with a real root at $x = a$ of multiplicity $r+1$ then $Q'(x)$ has r – fold roots at $x = a$. 7
- (ii) Solve the equation $x^4 - 5x^3 + 4x^2 + 3x + 9 = 0$ given that it has a root of multiplicity 2 over $\mathbb{C}$.
- c) If $z = \cos \theta + i \sin \theta$
- (i) Show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$ 4
- (ii) Hence by dividing throughout by z^2 or otherwise, solve the equation $5z^4 - 11z^3 + 16z^2 - 11z + 5 = 0$, given that $|z| = 1$. 2

Question 5 (15marks) START A NEW BOOKLET

- a) (i) Show that the equation of the tangent and normal at $P(a \cos \theta, b \sin \theta)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$ and $\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$ respectively. 3
- (ii) The tangent and normal at P cut the y – axis at A and B respectively. Find the coordinates of A and B. 2
- (iii) Show that the focus S lies on the circumference of the semi circle which has diameter AB. 3
- b) (i) Determine the real values of k for which $\frac{x^2}{4+k} + \frac{y^2}{9+k} = 1$ defines 3
- α) an ellipse
- β) an hyperbola
- (ii) If $k = -5$ in the above equation, find the eccentricity, the coordinates of the foci and the equations of the directrices of the conic. 2
- (iii) Draw a neat sketch of the conic indicating all key features. 2

Question 6 (15 marks) START A NEW BOOKLET

- a) Let $f(x) = \frac{4}{x} - x$. Provide separate half page sketches of the graphs of the following:

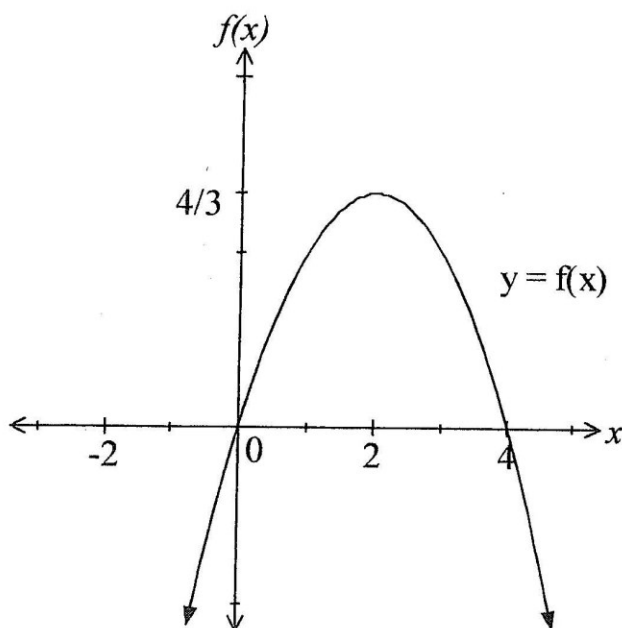
(i) $y = f(x)$ 2

(ii) $y = \sqrt{f(x)}$ 2

(iii) $y = e^{f(x)}$ 2

Label each graph carefully

b)



- (i) Use the diagram to find the values of a, b, c given $f(x) = ax^2 + bx + c$ 2

- (ii) Solve $-1 \leq f(x) \leq 1$ 3

- (iii) Hence or otherwise sketch 4

α $y = \ln[f(x)]$

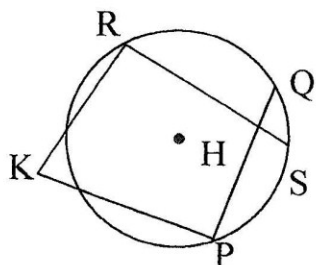
β $y = \cos^{-1}[f(x)]$

Question 7 (15 marks) START A NEW BOOKLET

- a) The base of a solid is a circular region of radius a units. Find the volume if every cross section of a plane perpendicular to a certain diameter is a square with one side lying in the base. 4
- b) Find, by the method of cylindrical shells, the volume of the solid generated when the region bounded by the curve $y = x^2 + 1$, the line $x = 2$ and the coordinate axes is rotated about the line $x = 3$. 5
- c) Find the value of x such that $\sin x = \cos 5x$ and $0 < x < \pi$ 6

Question 8 (15 marks) START A NEW BOOKLET

- a) PQ and RS are 2 chords of a circle. PQ and RS intersect at H. K is a point such that angle KPQ and angle KRS are right angles. Show that KH produced is perpendicular to QS.



- 6
- (b) A parachutist of mass m falls to ground from a plane. Given that air resistance is proportional to the square of his speed v : 9
- (i) Draw a diagram showing clearly the forces acting on the parachutist during his free fall.
- (ii) Deduce that $\frac{d}{dx}(v^2) = 2g - 2kv^2$
- (iii) Show that $v^2 = \frac{g}{k} - Ae^{-2kx}$ satisfies the differential equation in part (ii) and show that $A = \frac{g}{k}$
- (iv) Sketch the graph of v^2 against x and find an expression for the terminal speed of the parachutist during his free-fall.

End of Exam

Ascham
1999 4 unit Mathematics Trial - Solutions

①

1a) $\int 7x \sqrt{4x^2-3} dx = \frac{7}{8} (4x^2-3)^{3/2} \cdot \frac{1}{3} + C$
 $= \frac{7}{12} \sqrt{4x^2-3}^3 + C$

b) (i) $\int_0^{\sqrt{2}} \sqrt{4-x^2} dx = \int_0^{\pi/4} \sqrt{4 \cos^2 \theta} \cdot 2 \cos \theta d\theta$

let
 $x = 2 \sin \theta$
 $dx = 2 \cos \theta d\theta$
 when
 $x = \sqrt{2}, \theta = \pi/4$
 $x = 0 \theta = 0$

$= \int_0^{\pi/4} 4 \cos^2 \theta d\theta$
 $= 2 \int_0^{\pi/4} (1 + \cos 2\theta) d\theta$
 $= 2 \left[\frac{\sin 2\theta}{2} + \theta \right]_0^{\pi/4}$
 $= 1 + \frac{\pi}{2}$

(ii) $\int_0^{\pi} x \sin x = \int_0^{\pi} x \frac{d}{dx} (-\cos x) dx$
 $= \left[-x \cos x \right]_0^{\pi} + \int_0^{\pi} \cos x dx$
 $= \left[-x \cos x \right]_0^{\pi} + \left[\sin x \right]_0^{\pi}$
 $= \pi$

(iii) $\int_2^4 \frac{dx}{x^2-4x+8} = \int_2^4 \frac{dx}{(x-2)^2+4}$
 $= \frac{1}{2} \left[\tan^{-1} \left(\frac{x-2}{2} \right) \right]_2^4$
 $= \pi/8$

(iv) $\int_{-1}^1 \frac{4+x^2}{4-x^2} dx$

NA $\frac{4+x^2}{4-x^2} = -1 + \frac{8}{4-x^2}$

and $\frac{8}{4-x^2} = \frac{a}{2-x} + \frac{b}{2+x} \Rightarrow a=2, b=2$

$\therefore \int_{-1}^1 \frac{4+x^2}{4-x^2} dx = -\int_{-1}^1 dx + \int_{-1}^1 \frac{2dx}{2-x} + \int_{-1}^1 \frac{2dx}{2+x}$
 $= \left[-x - 2 \ln(2-x) + 2 \ln(2+x) \right]_{-1}^1$
 $= 4 \ln 3 - 2$

2a) $x^2 - 3ix + 4 = 0$
 $x = \frac{3i \pm \sqrt{-25}}{2}$

2(ii) let $12-5i = (x+iy)^2$ x, y
 $= x^2 - y^2 + 2ixy$

Thus $2xy = -5$ $x^2 - y^2 = 12$

Solving for x, y

$x^2 - \frac{25}{4x^2} = 12$

$4x^4 - 48x^2 - 25 = 0$

$x = \pm \frac{5}{\sqrt{2}}$ or $\pm \frac{i}{\sqrt{2}}$ but x, y real

$\therefore x = \pm \frac{5}{\sqrt{2}}, y = \mp \frac{1}{\sqrt{2}}$

$x+iy = \frac{1}{\sqrt{2}}(5-i)$ or $\frac{1}{\sqrt{2}}(-5+i)$

(iii) $z = \frac{u-i}{u-2}$ u is purely real

method 1. changing the subject to u

$u = \frac{2z-i}{z-1}$

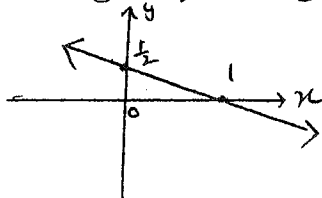
Since u is purely real, $\arg u = 0, \pi$

$\arg(2z-i) - \arg(z-1) = 0, \pi$

$\arg 2 + \arg(z - \frac{i}{2}) - \arg(z-1) = 0, \pi$

$\therefore \arg(z - \frac{i}{2}) = \arg(z-1)$ (Since a)

OR $\arg(z - \frac{i}{2}) = \arg(z-1) + \pi$



method 2 $u = \frac{2z-i}{z-1}$ let $z = x+iy$

$\therefore u = \frac{2(x+iy)-i}{x+iy-1} \Rightarrow \text{realise}$

$u = \frac{2x^2 + 2y^2 - 2x - 2y - i(2y + x - 1)}{(x-1)^2 + y^2}$

Since u is purely real, $2y+x-1=0$

Ascham 99 Total, Solutions.

5 b) (i) $\frac{x^2}{4+k} + \frac{y^2}{9+k} = 1$

x) For the curve to be an ellipse

$4+k > 0$ and $9+k > 0$

i.e. $k > -4$ and $k > -9 \Rightarrow k > -4$

β) For the curve to be a hyperbola

$4+k > 0$ & $9+k < 0$ OR $4+k < 0$ and $9+k > 0$

i.e. $k > -4$ & $k < -9$ OR $k < -4$ & $k > -9$
no solution $-9 < k < -4$

Thus for a hyperbola $-9 < k < -4$

(ii) If $k = -5$ the curve is a hyperbola

$\frac{y^2}{4} - \frac{x^2}{1} = 1$ (conjugate of $\frac{x^2}{1} - \frac{y^2}{4} = 1$)

To find e: let $a=2$ $b=1$

$b^2 = a^2(e^2 - 1)$

$e^2 = 5/4$

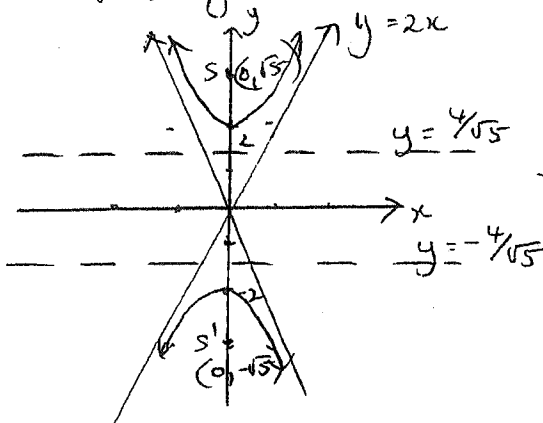
$e = \frac{\sqrt{5}}{2}$

foci $(0, \pm ae) = (0, \pm \sqrt{5})$

directrices: $y = \pm \frac{a}{e} = \pm \frac{4}{\sqrt{5}}$

asymptotes same as $\frac{x^2}{1} - \frac{y^2}{4} = 1$

$\therefore$ asymp. $y = \pm 2x$

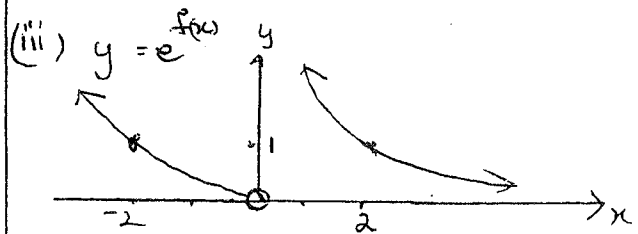
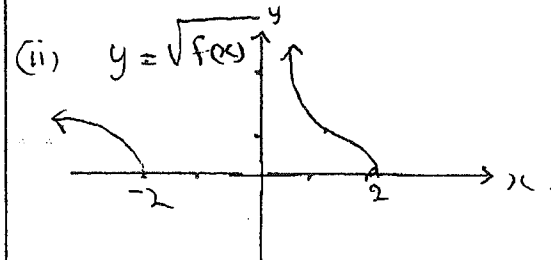
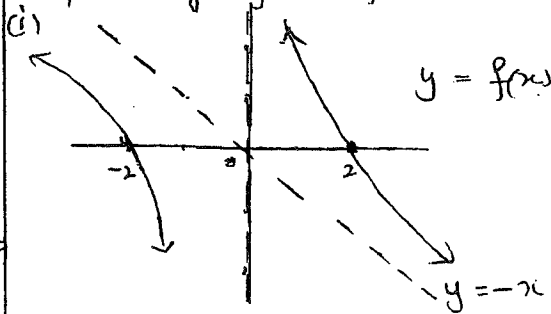


6. a) $f(x) = \frac{4}{x} - x$

• $\lim_{x \rightarrow \infty} f(x) = -x$, $\lim_{x \rightarrow 0} f(x) = \frac{4}{x}$

• intercepts: $x \neq 0$, $y = 0 \Rightarrow x = \pm 2$

• symmetry: $f(-x) = -f(x) \therefore$ odd fn.



b) (i) $y = ax^2 + bx + c$

$x=0$ $y=0 \Rightarrow c=0$ ①

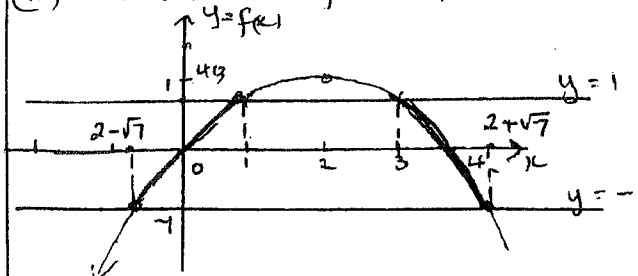
$x=4$ $y=0 \Rightarrow 16a + 4b = 0$ ②

$x=2$, $y=4/3 \Rightarrow \frac{2}{3} = 2a + b$ ③

$\therefore$ on soln: $a = -1/3$ $b = 4/3$

Thus $y = -\frac{x}{3}(x-4)$

(ii) soln to $-1 \leq f(x) \leq 1$

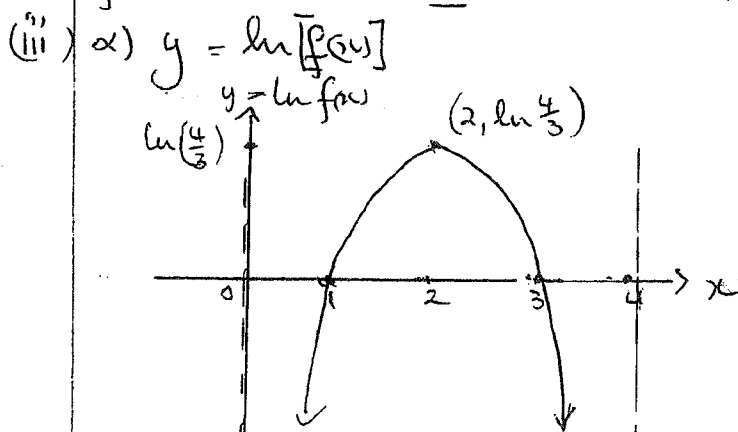


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444 Total Ascham 99

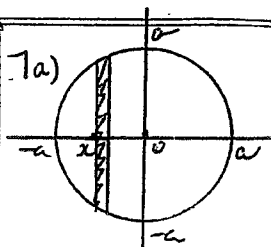
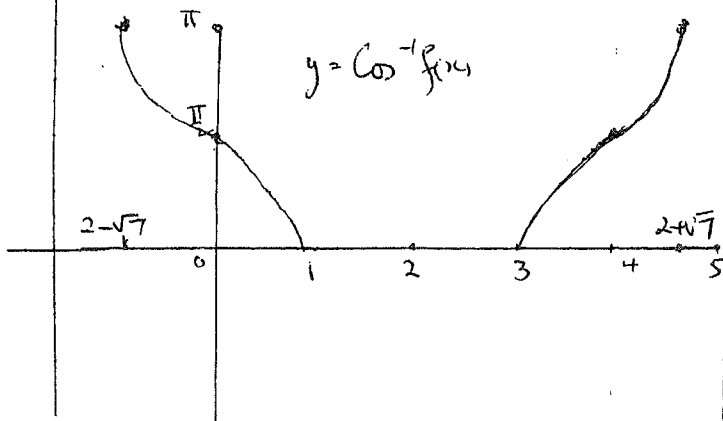
6 (ii) To solve $-1 \leq f(x) \leq 1$
 when $f(x) = 1$ $\frac{x}{3}(4-x) = 1$ $f(x) = -1$
 $\Rightarrow x = 1, 3$ $x^2 - 4x - 3 = 0$
 $x = 2 \pm \sqrt{7}$

Thus we see from the graph $y = f(x)$
 that $-1 \leq f(x) \leq 1$
 for $2 - \sqrt{7} \leq x \leq 1$ OR $3 \leq x \leq 2 + \sqrt{7}$



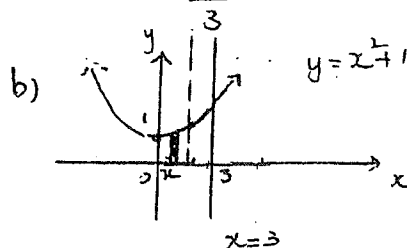
when $f(x) = 0$ $\ln f(x)$ is undefined.
 $f(x) = \frac{4}{3}$ $\ln f(x) = \ln \frac{4}{3}$
 $f(x) = 1$ $\ln f(x) = 0$

b) $y = \cos^{-1}[f(x)]$
 for $-1 \leq f(x) \leq 1 \Rightarrow 0 \leq \cos^{-1} f(x) \leq \pi$
 Thus for $2 - \sqrt{7} \leq x \leq 1$ $\cos^{-1} f(x) \leq \pi$
 $3 \leq x \leq 2 + \sqrt{7} \Rightarrow 0 \leq \cos^{-1} f(x) \leq \pi$
 check: $\cos^{-1} 0 = \frac{\pi}{2} \therefore y = \frac{\pi}{2}$ at $x = 0, 4$



typical $\delta V = [2 \text{ for } = 4y]$

Total $V = \int_{-a}^a 4(a^2 - x^2) dx$
 $= 8 \left[a^2 x - \frac{x^3}{3} \right]_0^a$
 $= \frac{16a^3}{3} u^3$



Let δV be the vol of a typical cylind shell at thickness δx

inner radius $= 3 - (x + \delta x)$

outer radius $= 3 - x$

$\delta V = \pi \{ (3-x)^2 - [3-(x+\delta x)]^2 \} y$
 $= \pi \{ 6\delta x - 2x\delta x - \delta x^2 \} y$
 $= \pi (6 - 2x) y \delta x$ $[\delta x^2]$
 $= \pi (6 - 2x) (x^2 + 1) \delta x$

Thus

$V = \int_0^2 (6x^2 - 2x^3 + 6 - 2x) dx$
 $= \left[2x^3 - \frac{x^4}{2} + 6x - x^2 \right]_0^2$
 $= 16\pi u^3$

c) $\sin x = \cos 5x$ $0 < x < \pi$

$\cos(\frac{\pi}{2} - x) = \cos 5x$

$\therefore \frac{\pi}{2} - x = 2n\pi \pm 5x$ (general formula)

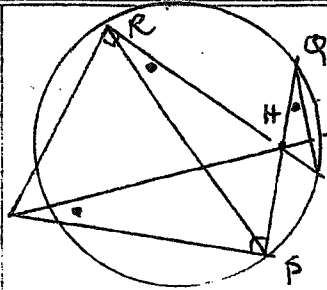
$\frac{\pi}{2} - x = 2n\pi + 5x$ OR $\frac{\pi}{2} - x = 2n\pi - 5x$

$x = \frac{1}{6}(\frac{\pi}{2} - 2n\pi)$ OR $x = \frac{1}{4}(2n\pi - \frac{\pi}{2})$

$n=0$ $x = \frac{\pi}{12}$, OR $n=1$, $x = \frac{3\pi}{8}$

$n=-1$ $x = \frac{5\pi}{12}$ $n=2$ $x = \frac{7\pi}{8}$

8a



Aim: to prove

$KH \perp QS$

Constr: join RP

Extend KH to meet QS at T

Now if KPTQ is a cyclic quad then
 $K\hat{P}Q = K\hat{T}Q = 90^\circ$

Proof: To show KPTQ is a cyclic quad

In Circle RQSP,

$S\hat{Q}P = S\hat{R}P$ (angles standing on same arc SP)

But $T\hat{R}P = S\hat{R}P$ (since KRP is cyclic opp L's = 90° , L's standing on same arc KP)

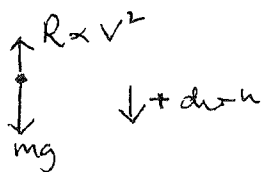
Thus $S\hat{Q}P = T\hat{R}P$ (angles equal to equal angles)

Thus KPTQ is a cyclic quad

$\therefore K\hat{P}Q = K\hat{T}Q = 90^\circ$

$\therefore KH$ prod. is $\perp QS$

b)



Let $R = mkv^2$ (choosing mk as constant of proportionality)

$$\therefore m\ddot{x} = mg - mkv^2$$

$$\ddot{x} = g - kv^2$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = g - kv^2$$

$$\frac{d}{dx} v^2 = 2g - 2kv^2$$

(iii) given $v^2 = \frac{g}{k} - Ae^{-2kx}$

$$\frac{d}{dx} v^2 = \frac{d}{dx} \left(\frac{g}{k} - Ae^{-2kx} \right)$$

$$\begin{aligned} \frac{d}{dx} v^2 &= 2Ake^{-2kx} \\ &= 2k \cdot Ae^{-2kx} \\ &= 2k \left(\frac{g}{k} - v^2 \right) \\ &= 2g - 2kv^2 \end{aligned}$$

Thus $v^2 = \frac{g}{k} - Ae^{-2kx}$ satisfies

$$\frac{d}{dx} v^2 = 2g - 2kv^2$$

Now when $x=0, v=0$

$$\therefore 0 = \frac{g}{k} - Ae^0$$

$$\therefore A = \frac{g}{k} \quad \therefore v^2 = \frac{g}{k} - \frac{g}{k} e^{-2kx} = \frac{g}{k} (1 - e^{-2kx})$$

(iv) $\lim_{x \rightarrow \infty} v^2 = \lim_{x \rightarrow \infty} \frac{g}{k} (1 - e^{-2kx}) = \frac{g}{k}$

Thus as $x \rightarrow \infty, v^2 \rightarrow \frac{g}{k},$
 $\rightarrow v \rightarrow \pm \sqrt{\frac{g}{k}}$

$\therefore$ terminal speed = $\sqrt{\frac{g}{k}}$

